

● NORMALIZATION QUESTION (MS)

StudentID (PK)	StudentName	DOB	Year	Activity	ActivityType	ActivityTimings	ActivityLead	ActivityJoiningDate	ActivityLevel	GuardianID	GuardianName	GuardianContact
101	Ammara Javed	13-02-2005	1st	Basketball	Sports	Morning	Mr. X	31-08-2024	Intermediate	1	Javed Jaffrey	555-1234
				Drama Club	Performance	Afternoon	Mr. Y		Beginner			
102	Bilal Syed	22-08-2004	2nd	Chess Club	Hobby	Evening	Ms. A	15-09-2024	Beginner	2	Mariam Syed	555-5678
				Coding	Educational	Morning	Mr. B	10-09-2023	Advanced			
				Debate	Performance	Afternoon	Mr. C	01-01-2024	Intermediate			
103	Dawood Waqar	10-09-2005	1st	Football	Sports	Evening	Ms. D	15-06-2024	Beginner	3	Sarah Bilal	555-4321
				Robotics	Educational	Morning	Mr. E	10-08-2024	Advanced			
				Robotics	Educational	Afternoon	Mr. F	01-09-2024	Beginner			
104	Erum Pervez	20-01-2002	4th	Drama Club	Performance	Morning	Mr. G	10-08-2024	Advanced	4	Pervez Waheed	555-9876
				Cooking Club	Hobby	Afternoon	Ms. A	01-09-2021	Advanced			
105	Faheem Haq	23-05-2003	3rd	Debate	Performance	Evening	Ms. J	19-01-2024	Beginner	5	Nadeem Haq	555-6543
				Basketball	Sports	Afternoon	Ms. G	04-12-2023	Beginner			
				Coding	Educational	Morning	Mr. X	22-11-2022	Advanced			
106	Ghazal Khan	24-07-2004	2nd	Chess Club	Hobby	Afternoon	Mr. Y	07-10-2022	Advanced	6	Mikaal Khan	555-3456
				Robotics	Educational	Evening	Ms. D	26-10-2023	Intermediate			
				Cooking Club	Hobby	Morning	Mr. B	19-06-2024	Beginner			
107	Harish Chandra	01-01-2022	4th	Football	Sports	Evening	Mr. I	29-04-2024	Beginner	7	Laxman Chandra	555-7890
				Football	Sports	Morning	Mr. E	18-03-2021	Advanced			
108	Ibrahim Shah	21-04-2003	3rd	Basketball	Sports	Afternoon	Mr. F	12-12-2022	Intermediate	8	Adam Shah	555-6789
				Basketball	Sports	Morning	Mr. X	12-12-2022	Intermediate			
109	Javeria Merchant	20-09-2004	2nd	Cooking Club	Hobby	Afternoon	Mr. Y	17-11-2023	Intermediate	9	Ehtasham Merchant	555-1239
				Cooking Club	Hobby	Afternoon	Ms. J	17-11-2023	Intermediate			
110	Kiran Zahid	29-03-2005	1st	Coding	Educational	Afternoon	Mr. C	30-08-2024	Beginner	10	Zahid Siddiqui	555-0123
				Drama Club	Performance	Evening	Ms. D	30-08-2024	Beginner			
				Drama Club	Performance	Evening	Ms. A	30-08-2024	Beginner			
111	Monis Shahmeer	16-04-2003	3rd	Coding	Educational	Afternoon	Mr. C	15-06-2023	Intermediate	11	Shahmeer Zulfiqar	555-2345
				Coding	Educational	Evening	Ms. D	15-06-2023	Intermediate			
				Football	Sports	Morning	Mr. E	18-12-2022	Advanced			
112	Abdul Ahad	13-05-2002	4th	Football	Sports	Afternoon	Mr. F	25-08-2021	Advanced	12	Abdul Samad	555-4567
				Basketball	Sports	Morning	Mr. X	27-08-2021	Advanced			
				Debate	Performance	Afternoon	Mr. Y	01-01-2022	Intermediate			

Imagine you are employed as a Data Analyst at Habib University responsible for managing students' extracurricular activities information. A student can be a part of multiple extracurricular activities. Each of these activities are scheduled at different timings. During each time slot, there is always an Activity Lead who overlooks all the participating students in that group. A student can attend any of the time slots, as per their availability, so no specific slot will always have the same set of students. The university administration is using a table to store this information. Some of the rows from that table are given in Table 1 for reference.

1. Please analyze the provided extracurricular activities data table, and derive a relation schema for it in 1NF, by specifying the primary keys. Give reasons for choosing the primary keys.
2. Identify full, partial, and transitive functional dependencies that exist among its attributes. If there are no dependencies, please state that explicitly.
3. Normalize the schema up to the Third Normal Form (3NF) by creating new tables. Provide schemas for 2NF, and 3NF.

Solution

1NF:

Extracurricular (StudentID, StudentName, DOB, Year, Activity, ActivityType, ActivityTimings, ActivityLead, ActivityJoiningDate, ActivityLevel, GuardianID, GuardianName, GuardianContact)

2NF:

{StudentID} → {StudentName, DOB, Year, Guardian, GuardianName, GuardianContact}
{Activity} → {ActivityType}
{StudentID, Activity} → {ActivityJoiningDate, ActivityLevel}
{Activity, ActivityTimings} → {ActivityLead}

Students (StudentID, StudentName, DOB, Year, GuardianID, GuardianName, GuardianContact)

Activities (Activity, ActivityType)

StudentActivity (StudentID, Activity, ActivityJoiningDate, ActivityLevel)

ActivityLeads (Activity, ActivityTimings, ActivityLead)

StudentExtracurricular (StudentID, Activity, ActivityTimings)

3NF:

{StudentID} → {GuardianID} → {GuardianName, GuardianContact}

Students (StudentID, StudentName, DOB, Year, GuardianID)

Guardians (GuardianID, GuardianName, GuardianContact)

Activities (Activity, ActivityType)

StudentActivity (StudentID, Activity, ActivityJoiningDate, ActivityLevel)

ActivityLeads (Activity, ActivityTimings, ActivityLead)

StudentExtracurricular (StudentID, Activity, ActivityTimings)

- **ERD QUESTION (AT)**

A fruit orchard company manages multiple orchards across different regions of Pakistan. Each orchard specializes in cultivating a variety of fruits (e.g., mango, orange, apple, guava). Each orchard has a location (city, province) and is managed by a designated orchard manager.

Every orchard must hold a government-issued license to operate. Each orchard has exactly one license, and each license belongs to exactly one orchard. For each license, the system records the license ID, issue date, and expiry date.

Every orchard is divided into multiple plots. Each plot is used to cultivate exactly one type of fruit, but the same fruit may be grown in different plots (sometimes even within the same orchard). Each plot has information about its soil type, area (in acres), and year it was first cultivated.

Each fruit type has details such as fruit name, season, and average yield per acre. Fruits are harvested multiple times in a year. For each harvest, the system keeps track of the date, the quantity harvested (in kg), and the plot it came from.

The orchard company employs multiple workers. Each worker is assigned to one orchard but may be rotated between different plots within that orchard. Workers can be specialized (e.g., irrigation, pesticide application, harvesting), and the system stores each worker's name, CNIC, contact, and specialization. Workers are supervised by orchard managers, but an orchard manager may supervise multiple orchards.

The company sells fruits in bulk to different buyers (wholesalers, retailers, or exporters). For each sale, the system stores the buyer details (name, type, contact), the fruit type sold, the quantity sold, the price per kg, and the date of sale. Sometimes, a single sale can include multiple fruit types.

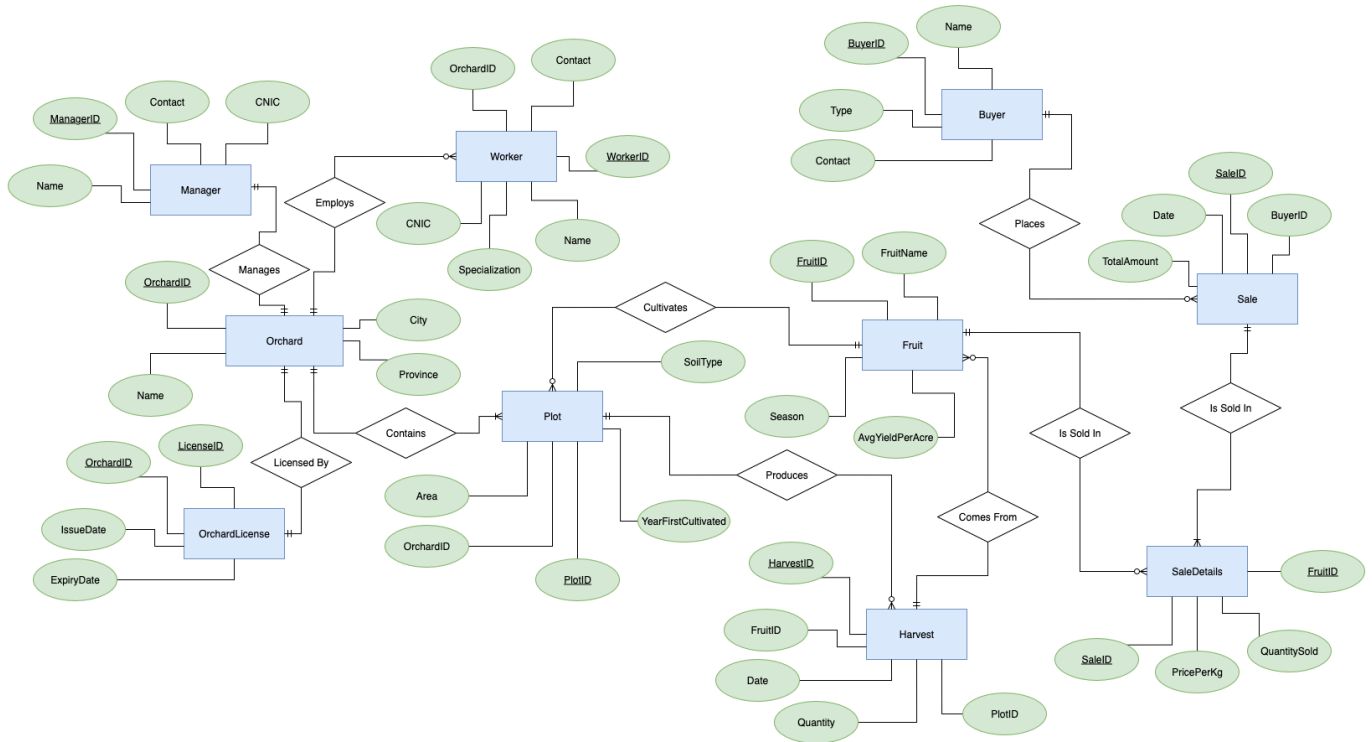
Management is interested in generating the following reports:

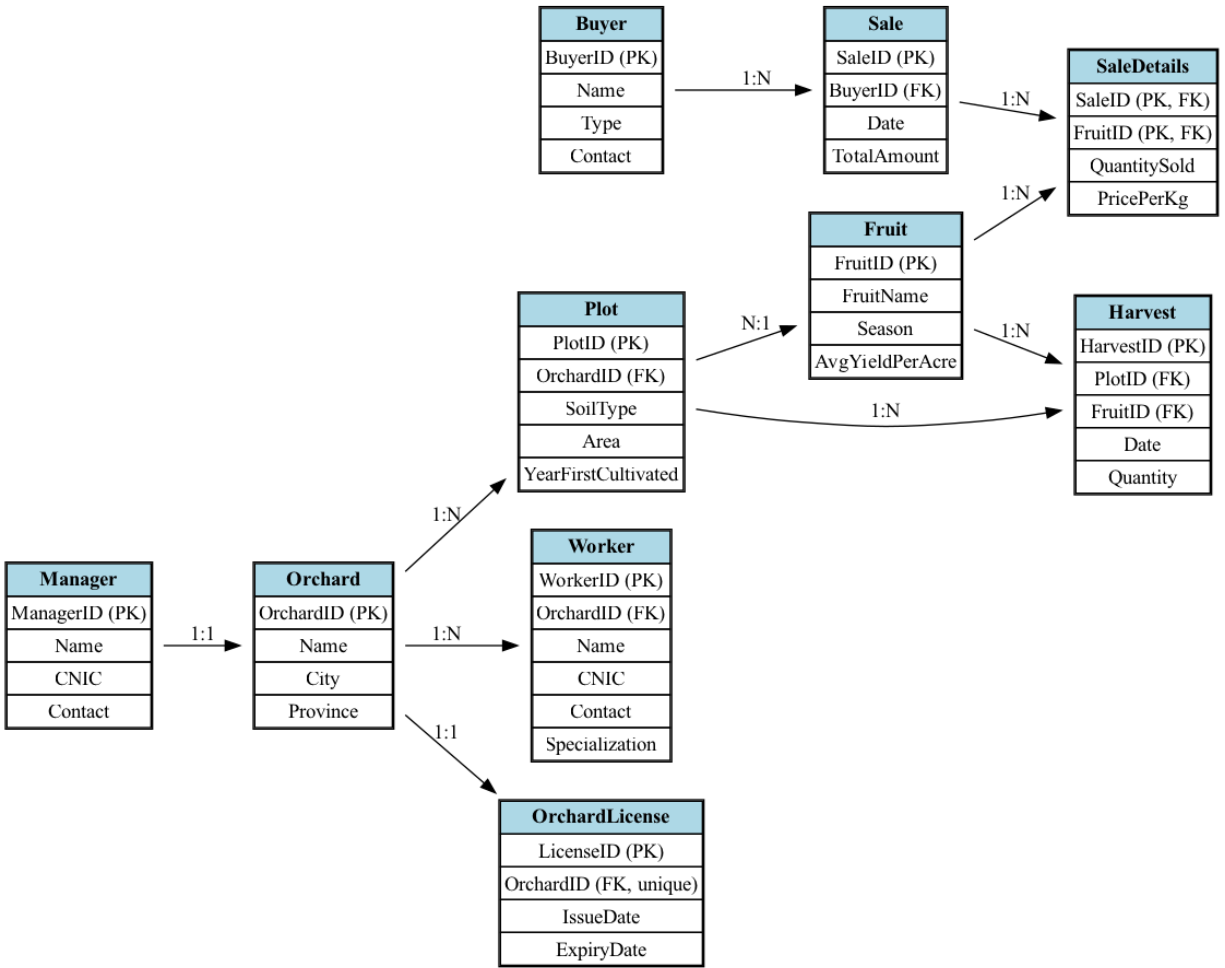
- Annual yield per fruit type per orchard
- Which orchard produces the most fruit of a given type
- Productivity of workers in terms of harvested quantity
- Sales trends of different fruit types across years
- The contribution of each orchard to total revenue
- Expiring orchard licenses in the upcoming year

Based on the above scenario, construct an Entity-Relationship Diagram (ERD) that includes:

- Entities with appropriate attributes (identify primary keys and foreign keys).
- Relationships among entities with appropriate cardinalities.
- Any assumptions you make to clarify ambiguities in the scenario.

Solution





- **Khaadi ERD QUESTION (UT)**

Khaadi is one of the most popular fashion brands in Pakistan. You have been assigned the task of designing a data model for their online product catalog. The business details are provided below. Your task is to construct a normalized data model that includes:

- Entities, attributes, primary keys (PKs), and foreign keys (FKs)
- Relationships and their cardinalities

Business Details

Khaadi's catalog is organized into various departments as is evident from their website:



Each department contains multiple categories and sub-categories. A category can belong to multiple departments. A sub-category can belong to multiple categories.

Unstitched		Ready To Wear		Kids	Accessories	<u>Home</u>	Beauty
Bedroom	Bathroom	Dining Room	Cushion Covers		Kids Collection		
Bed Covers	Shower Curtain	Table Covers	Digital Printed		Bed Sheets		
Bed Sheets	Towels	Table Runners	Embroidered		Cushion Covers		
Duvet Covers		Placemats	Handwoven		Duvet Covers		
Foot Mats		Hot Plates			Fitted Sheet		
Floor Runners		Coasters			Towels		
Pillow Covers		Roti Cloth					
Placemats							
Fitted Sheets							





Embroidered Kurta

PKR 1,824

SKU: BXZ194015-Blue



Size

2-3

3-to-4

4-5

5-6

6-to-7

7-to-8

8-to-9

9-to-10

11-to-12

10-to-11



Quantity

-

1

+



ADD TO CART

SIZE CHART

Order by

Share

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Description

Embroidered Kurta

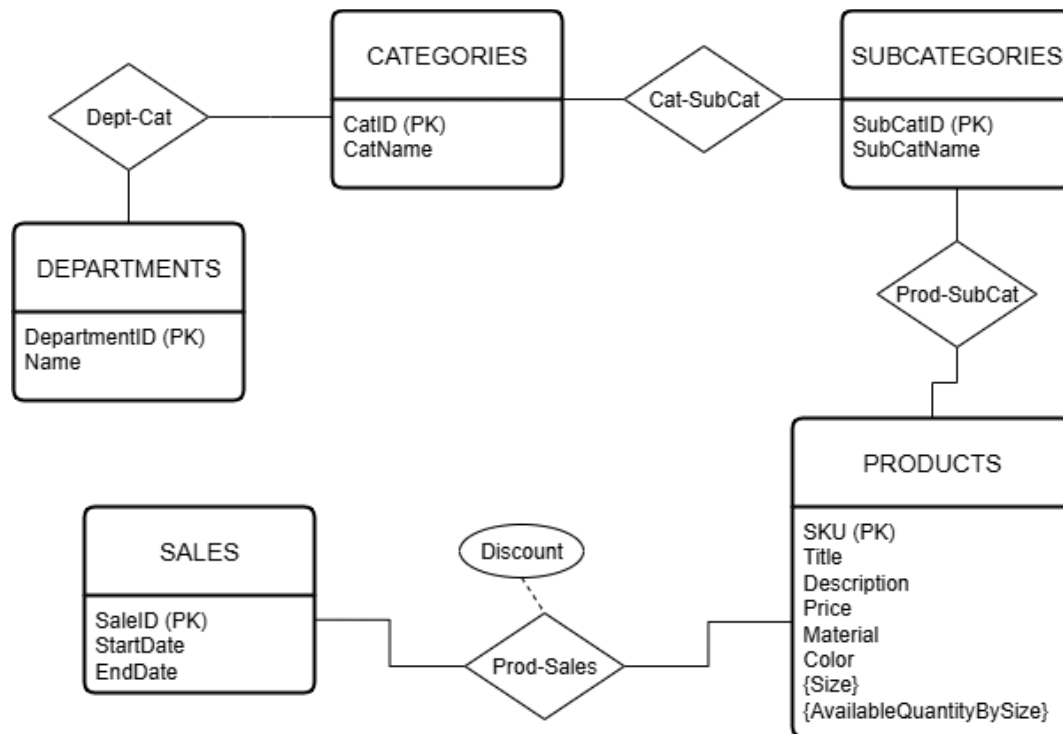
Details

Material	Khaddar
Color	Blue

Throughout the year, different sales events are launched. During these sales, selected products are offered at discounted prices. To set up a sale, the system specifies a timeline (start and end dates), the products included in the sale, and the percentage discount applied to each product. Not all products are required to be part of a sale, and different products may have different discount percentages.

Solution:

ERD:



Relation Schema:

